Chemistry • Nucleus, electrons and isotopes

Atomic Structure

Official exercise • Complete solved answers • Verified isotope calculations

Chapter 2 • Complete official exercise • 10 MCQs + 5 short + 5 constructed + 4 descriptive + 2 investigative

What this chapter is about

• **Atoms** contain a tiny, dense nucleus of protons and neutrons, surrounded by electrons. Atomic number Z identifies an element; mass number A counts protons plus neutrons.

• **Bohr's model** uses fixed energy levels to explain electron transitions. The modern quantum model describes orbitals and probabilities rather than exact planetary tracks.

• **Isotopes** have the same number of protons but different numbers of neutrons. They have nearly identical chemical behaviour but different masses and some physical properties.

• **Radioactive isotopes** have unstable nuclei and emit radiation. Their applications include imaging, treatment, dating and industrial measurements.

Exercise coverage: All five official sections are solved in book order. Isotope symbols and numerical data were checked against the rendered textbook pages.

studysapiens-

A. Multiple Choice Questions

1. How many electrons can be accommodated at the most in the third shell of the elements?

(a) 8 (b) 18
(c) 10 (d) 32

Answer: (b) 18 — For shell n = 3, maximum capacity = $2n^2 = 2 \times 3^2 = 18$ electrons.

2. What information was obtained from discharge tube experiments?

(a) structure of atom was discovered (b) neutrons and protons were discovered
(c) electrons and protons were discovered (d) presence of nucleus in an atom was discovered

Answer: (c) electrons and protons were discovered — Cathode-ray and canal-ray experiments provided evidence for electrons and positive particles, including the proton.

3. Why have isotopes not been shown in the periodic table?

(a) periodic table cannot accommodate a large number of isotopes of different elements
(b) some of the isotopes are unstable and they give rise to different elements
(c) all the isotopes have same atomic number; so there is no need to give them separate places
(d) isotopes do not show periodic behavior

Answer: (c) all the isotopes have same atomic number; so there is no need to give them separate places — Isotopes of an element have the same proton number Z and therefore occupy the same element position in the periodic table.

4. Which particle is present in different number in the isotopes?

(a) electron
(b) neutron
(c) proton
(d) both neutron and electron

Answer: (b) neutron — Isotopes have equal proton numbers but different neutron numbers; neutral atoms also have equal electron numbers.

5. In which isotope of oxygen are there equal numbers of protons, electrons and neutrons?

(a) ^{17}O
(b) ^{16}O
(c) ^{18}O
(d) none of these

Answer: (b) ^{16}O — Neutral ^{16}O has 8 protons, 8 electrons and $16 - 8 = 8$ neutrons.

6. What will be the relative atomic mass of nitrogen given the abundances of its two isotopes, ^{14}N and ^{15}N, as 99.64 and 0.35 respectively?

(a) 14.0210
(b) 14.0021
(c) 14.2100
(d) 14.1200

Answer: (b) 14.0021 — Using the supplied percentages: $A_r = (14 \times 99.64 + 15 \times 0.35)/100 = 14.0021$.

7. How is radiocarbon dating useful for archaeologists?

(a) it helps determine the age of organic matter
(b) it helps determine the composition of matter
(c) it helps determine the usefulness of matter
(d) it helps determine whether the matter is radioactive or not

Answer: (a) it helps determine the age of organic matter — The remaining carbon-14 in once-living material is used to estimate the time since the organism died.

studysapiens.com

8. What keeps the particles present in the nucleus intact?

(a) particles are held together by strong nuclear force
(b) particles are held together by weak nuclear force
(c) particles are held together by electrostatic force
(d) particles are held together by dipolar force

Answer: (a) particles are held together by strong nuclear force — The short-range strong nuclear attraction between nucleons overcomes proton–proton electrostatic repulsion at nuclear distances.

9. How do electrons keep themselves away from the oppositely charged nucleus?

(a) by keeping themselves stationary
(b) by revolving around the nucleus
(c) due to their wave-like nature
(d) a magnetic field around the nucleus keeps them away

Answer: (b) by revolving around the nucleus — In the Bohr picture intended here, electrons revolve in allowed energy levels. The modern quantum model replaces exact paths with probability orbitals.

10. Rubidium consists of two isotopes ^{85}Rb and ^{87}Rb. The percent abundance of the light isotope is 72.2%. What is the percent abundance of the heavier isotope? Its atomic mass is 85.47.

(a) 15%
(b) 27.8%
(c) 37%
(d) 72%

Answer: (b) 27.8% — For two isotopes, abundances total 100%: heavier isotope = 100 − 72.2 = 27.8%.

B. Questions for Short Answers

2.1. Why is it said that almost all the mass of an atom is concentrated in its nucleus?

Protons and neutrons are each about 1,836 times as massive as an electron and both are located in the nucleus. Electrons occupy almost all the atomic volume but contribute very little mass; hence more than 99.9% of the mass lies in the tiny nucleus.

2.2. Why are elements different from one another?

Each element has a unique **atomic number**, meaning a characteristic number of protons. Neutral atoms therefore have different electron numbers and arrangements, which produce different chemical and physical properties.

2.3 How many neutrons are present in $^{210}_{83}$Bi?

Given: Mass number A = 210; atomic number Z = 83.

studysapiens.com

$$N = A - Z$$

$$N = 210 - 83 = 127$$

Answer: 127 neutrons

2.4. Why is tritium ($^{3}_{1}H$) a radioactive element?

Tritium has one proton and two neutrons. Its nucleus is unstable and emits radiation to reach a more stable state; tritium undergoes beta decay and forms helium-3. It is the nucleus—not the electron arrangement—that makes it radioactive.

2.5. How can an atom absorb and evolve (emit) energy?

An atom absorbs energy when an electron moves from a lower allowed energy level to a higher one. The atom emits energy, often as a photon, when the electron returns to a lower level. The absorbed or emitted energy equals the difference between the two levels: $\mathbf{\Delta E = E_{higher} - E_{lower}}$.

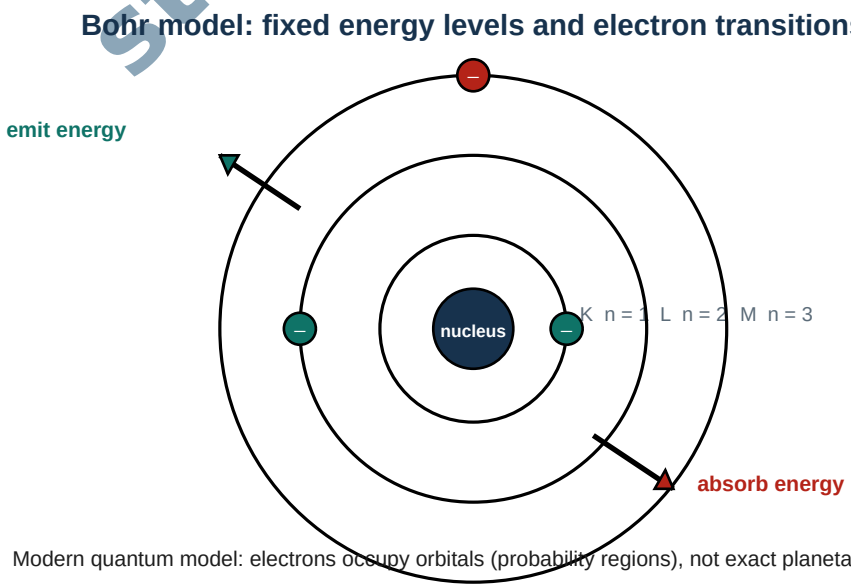
C. Constructed Response Questions

2.1. Why does the energy of an electron increase as we move from the first shell to the second shell?

The second shell has a greater average distance from the positively charged nucleus. An electron there is less strongly bound and has higher potential energy than in the first shell.

In the Bohr model the allowed levels are quantized, so the second shell corresponds to a specific higher energy: $\mathbf{E_2 > E_1}$. Moving an electron from K to L therefore requires absorption of exactly the energy difference.

Figure 1 — Original energy-level and transition diagram



Modern quantum model: electrons occupy orbitals (probability regions), not exact planetary paths.

studysapiens.com

2.2. Why is it needed to lower the pressure of the gas inside the discharge tube?

At low pressure there are fewer gas molecules per unit volume, so charged particles can travel longer distances without frequent collisions. This allows a sustained electrical discharge at high voltage and lets cathode or canal rays cross the tube with little scattering, making their glow and deflection observable.

2.3. What is the classical concept of an electron? How has this concept changed with time?

Classically, and in Bohr's early model, an electron was pictured as a tiny negatively charged particle revolving around the nucleus in a definite circular path, somewhat like a planet around the Sun. In the modern quantum model it is a quantum particle with wave-like behaviour. We describe an **orbital**, a region where there is a high probability of finding it; the uncertainty principle means its exact path and location cannot be mapped simultaneously with complete certainty.

2.4. Why are the nuclei of radioactive elements unstable?

Their proton–neutron composition does not give a stable balance between the short-range strong nuclear attraction and electrostatic repulsion among protons, or the nucleus contains excess energy. Such nuclei emit alpha, beta or gamma radiation and change into more stable nuclei; this spontaneous transformation is radioactive decay.

2.5. During discharge tube experiments, how did the scientists conclude that the same type of electrons and protons are present in all the elements?

Cathode rays had the same negative charge, deflection behaviour and charge-to-mass characteristics regardless of the cathode material or gas, showing that electrons are universal constituents of atoms. Positive canal rays depended on the gas, identifying positive ions of that gas; Rutherford's later experiments showed that the hydrogen nucleus—the **proton**—is a fundamental component of other nuclei. Thus electrons and protons are common building blocks, while their numbers differ between elements.

D. Descriptive Questions

2.1. Explain the structure of a hydrogen atom.

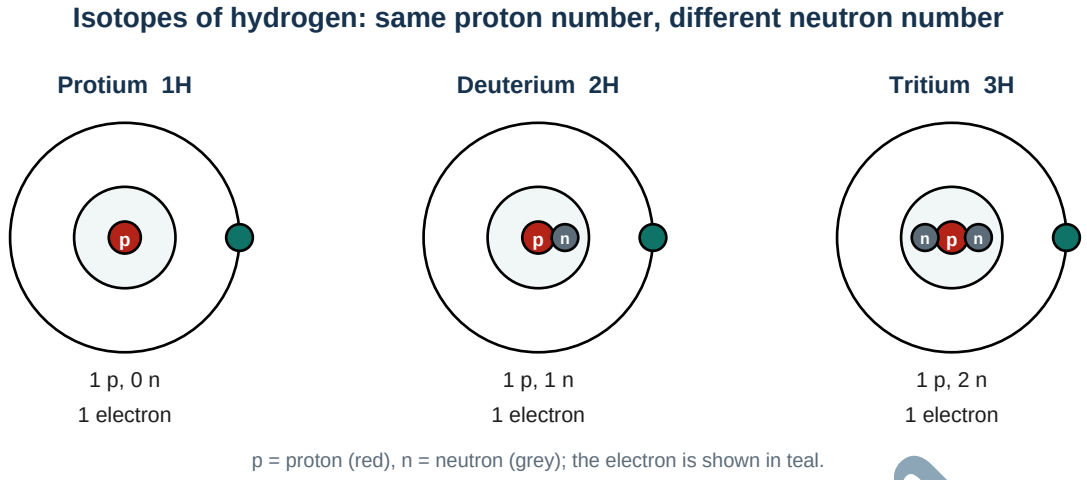
The ordinary hydrogen atom, ${}^{1}_{1}\mathrm{H}$, has a nucleus containing **one proton and no neutron**. One electron occupies the region around the nucleus; in the Bohr description it is in the K shell (n = 1).

The proton charge is +1 and the electron charge is −1, so the atom is electrically neutral. The proton contributes almost all the mass, while the electron occupies most of the atomic volume.

studysapiens.com

Hydrogen also has two other isotopes: **deuterium** ($^{2}_{1}H$; 1 p, 1 n, 1 e^{-}) and radioactive **tritium** ($^{3}_{1}H$; 1 p, 2 n, 1 e^{-}).

Figure 2 — Original comparison of hydrogen isotopes



p = proton (red), n = neutron (grey); the electron is shown in teal.

2.2. How does the theory of atomic structure explain the ionization of atoms by a radioactive isotope?

A radioactive isotope emits energetic radiation. If this radiation transfers enough energy to an atom, one or more electrons can overcome the attraction of the nucleus and escape.

$$Na + \text{ionizing energy} \rightarrow Na^{+} + e^{-}$$

The remaining atom has more protons than electrons and becomes a positive ion (cation). The ejected electron may attach to another atom or molecule, producing a negative ion. Alpha and beta particles and gamma photons can produce such ionization, although their penetrating and ionizing behaviours differ.

2.3. What is radioactivity? Explain any three applications of radioactive isotopes.

Radioactivity is the spontaneous emission of radiation by unstable nuclei as they decay towards more stable forms. The isotope emitting the radiation is a radioactive isotope (radioisotope).

1. Medical imaging: A small, controlled amount of technetium-99m is introduced into a patient. A special camera detects its gamma emissions and shows how the tracer moves through organs, helping diagnosis.

2. Radiocarbon dating: Living organisms maintain a nearly constant carbon-14 proportion. After death, carbon-14 decreases by decay; measuring the remaining proportion in wood, bone or other organic material estimates its age.

3. Cancer treatment: Carefully directed radiation, for example from cobalt-60 or other clinical sources, damages rapidly dividing cancer cells. The dose and targeting are controlled to limit harm to healthy tissue.

Other uses include thickness and strength testing, leak tracing, sterilization and nuclear power. Every use requires suitable shielding, monitoring and dose control.

studysapiens.com

2.4. Find out the relative atomic mass of mercury from the following data: ^{196}Hg = 0.0146%, ^{198}Hg = 10.02%, ^{199}Hg = 16.34%, ^{200}Hg = 23.13%, ^{201}Hg = 13.22%, ^{202}Hg = 29.80%, ^{204}Hg = 6.85%.

$$A_r = \Sigma(\text{relative isotopic mass} \times \text{percentage abundance}) / 100$$

$$\Sigma(m \times p) = 196(0.0146) + 198(10.02) + 199(16.34)$$

$$+ 200(23.13) + 201(13.22) + 202(29.80) + 204(6.85)$$

$$\Sigma(m \times p) = 2.8616 + 1983.96 + 3251.66 + 4626 + 2657.22 + 6019.60 + 1397.40$$

$$\Sigma(m \times p) = 19938.7016$$

$$\mathbf{A_r = 19938.7016 / 100 = 199.387016 \approx 199.39\ amu}$$

Data check: The percentages printed in the exercise total **99.3746%**, not 100%. The result above follows the textbook formula exactly. If the same listed data are first normalized to 100%, the weighted mean is **200.64 amu**; this difference arises from the supplied abundance data, not from arithmetic.

E. Investigative Questions

2.1. How can scientists synthesize elements in the laboratory?

Scientists synthesize new elements through **nuclear reactions**, not ordinary chemical reactions. A target nucleus is bombarded with neutrons, protons, alpha particles or heavier nuclei in a nuclear reactor or particle accelerator.

If the projectile fuses with or otherwise changes the target nucleus, the proton number Z changes. Because Z identifies an element, a new element or isotope can form. Scientists detect its characteristic radiation and decay products to confirm its identity.

Example: $^{249}_{98}\text{Cf} + {}^{48}_{20}\text{Ca} \rightarrow {}^{294}_{118}\text{Og} + 3\ {}^{1}_{0}\text{n}$

Many synthetic nuclei are extremely unstable and exist only briefly, so sensitive detectors and repeated independent measurements are essential.

2.2. A system just like our solar system exists in an atom. Comment on this statement.

The statement is a useful but limited analogy. In both pictures, a small central body contains most of the mass and much smaller objects occupy a much larger surrounding region. Rutherford’s nuclear atom and Bohr’s early model therefore invited comparison with the Sun and planets.

studysapiens.co

Feature	Solar system	Modern atom
Dominant attraction	Gravity	Electrostatic attraction between nucleus and electrons
Paths	Approximately definite elliptical orbits	No exact trajectories; orbitals describe probability distributions
Energy	Classical, effectively continuous changes	Quantized electronic energy levels
Description	Classical mechanics	Quantum mechanics and the uncertainty principle

Thus the “miniature solar system” is a helpful introductory model for the nucleus and extranuclear region, but it is **not a literal modern description** of electron motion.

Quick Revision: Particle Counts

Use A and Z first; then adjust only the electron count for ionic charge.

Species	Protons	Neutrons	Electrons	Particle-count check
$^{23}_{11}Na$	11	12	11	neutral: p = e
$^{35}_{17}Cl$	17	18	17	n = 35 − 17
$^{40}_{18}Ar$	18	22	18	n = 40 − 18
$^{23}_{11}Na^{+}$	11	12	10	one electron lost
$^{35}_{17}Cl^{-}$	17	18	18	one electron gained

Mass reminder: changing neutrons produces another isotope; changing protons produces another element; changing electrons produces an ion.

studysapiens.com

Points to Remember

Atomic number Z = number of protons; **mass number A** = protons + neutrons; **N = A − Z**.

A neutral atom has equal numbers of protons and electrons. Losing electrons forms a cation; gaining electrons forms an anion.

Maximum shell capacity in the simple model is $\mathbf{2n^2}$: K = 2, L = 8, M = 18, N = 32.

The strong nuclear force binds nucleons at very short distances.

Electrons absorb energy on moving to higher levels and emit energy on returning to lower levels.

Isotopes have the same Z but different N and A. Their chemical properties are nearly the same, while mass-dependent physical properties may differ.

Relative atomic mass is the abundance-weighted mean isotopic mass on the carbon-12 scale.

Radioactivity is a nuclear change and may alter the proton number and hence the identity of the element.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	c	3	c	4	b	5	b
6	b	7	a	8	a	9	b	10	b

How to attempt this chapter in the paper

• Write A and Z before calculating particles: p = Z, e = Z in a neutral atom, and n = A − Z.
• For isotope averages, show every product and divide by 100 only when percentages are normalized to total 100%; check the data first.
• Distinguish nuclear change from chemical change: radioactivity involves the nucleus, whereas ordinary bonding involves electrons.
• When asked about Bohr's model, state its usefulness and then mention that modern orbitals replace definite electron paths.